
Systematic analysis of on-premise and cloud services

Asif Ali, Asif Ali Laghari* and
Irfan Ahmed Kandhro

Department of Computer Science,
Sindh Madressatul Islam University,
Karachi, Pakistan
Email: asifali.telecom@gmail.com
Email: asif.laghari@smiu.edu.pk
Email: Irfan@smiu.edu.pk
*Corresponding author

Kamlesh Kumar

Department of Software Engineering,
Sindh Madressatul Islam University,
Karachi, Pakistan
Email: Kamlesh@smiu.edu.pk,

Salman Younus

Department of Computer Science,
Sindh Madressatul Islam University,
Karachi, Pakistan
Email: salmanyounus@gmail.com

Abstract: There are two key distinctions between cloud and on-premise (OP) software, the cost for each varies and so does the level of control. As organisations explore to reduce costs, many data and rules are migrating to multiple clouds like GCP, AWS, AZURE, and so on. Cloud service providers provide the extensibility, resilience, and quickness that traditional OP deployments usually lose. This paper presents a comparative analysis of on-premise and cloud computing differs by outline, arrangement, administration, and devices for associations and clients. This comparison shows that cloud computing provides more flexible infrastructure and better service of data processing on-premise. In today's enterprise IT world, there are many factors that a business must consider when deciding whether a cloud infrastructure is the right choice. As there are many mid-level organisations that I personally know which fails to adapt cloud solutions, instead rely on their proven legacy and on-premises applications and software to do business. Therefore, it is very challenging to decide for business, if either a cloud infrastructure is right choice or not. In this article, which have covered many areas and analyses of on-premise and cloud, which will help mid-level organisations to make the decision of choosing cloud solutions for their running business applications.

Keywords: key differences; cloud computing; on-premise pros and cons; cloud pros and cons; on-premise risk; security; cloud services provider selection; cost; compliance; cloud deployments methods.

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Biographical notes: Asif Ali is a highly accomplished professional with a solid educational background and extensive experience in the IT industry. He obtained his Bachelor's degree in Telecommunication Technology from Sindh University Jamshoro, Hyderabad, Sindh, Pakistan, in 2007. In pursuit of further knowledge and expertise, he recently completed his Master's degree in Computer Science from Sindh Madressatul Islam University, Pakistan in 2023. With over 13 years of dedicated service in the IT industry. He has consistently excelled in his role as a consultant. Throughout his career, he has been actively involved in delivering exceptional IT services to various organisations.

Asif Ali Laghari received his BS degree in Information Technology from the Quaid-e-Awam University of Engineering Science and Technology Nawabshah, Pakistan, in 2007 and Master's degree in Information Technology from the QUEST Nawabshah Pakistan in 2014. From 2007 to 2008, he was a Lecturer in the Computer and Information Science Department, Digital Institute of Information Technology, Pakistan. In 2015, he joined the school of the Computer Science and Technology, Harbin Institute of Technology, where he is now a PhD student. He has published more than ten technical articles in scientific journals and conference proceedings. His current research interests include Computer networks, cloud computing, and multimedia QoE management.

Irfan Ahmed Kandhro earned his Master of Science in Computer Science from Mohammad Ali Jinnah University Karachi, Pakistan in 2019. He is pursuing his PhD in Computer Science from Sindh Madressatul Islam University Karachi, Pakistan. He has served for more than seven years as a Lecturer and Software Engineer. He has published multiple research papers in international and ISI indexed journals, conferences, and workshops. His areas of interest are programming languages machine learning, deep learning, computer vision, and natural language processing.

Kamlesh Kumar received his PhD in Computer Science and Technology from The University of Electronic Science & Technology of China (UESTC) in 2016. He has more than ten years of teaching experience in various academic institutes. He has 20 research papers to his credit presented in various national/international conferences and published in HEC recognised journals. He has also been honoured with the award for Best Academic Research by UESTC in 2015. His research interests include image processing, computer vision, machine learning, knowledge discovery and data mining, and content based image retrieval system.

Salman Younus earned his Engineering degree in Telecommunication from Hamdard University, Karachi Pakistan in 2010. He is pursuing his Masters's degree in Computer Science from Sindh Madersatul Islam University Karachi Pakistan. He is serving in IT industry from last 13 years and currently working as a Cloud/DevOps administration consultant in one of the multi national organisation. He completed many projects purely implemented on Cloud and

DevOps technology. His areas of interests include architect cloud solution, DevOps pipeline to automate deployments and fetching code without human interventions

1 Introduction

Overall the IT industry technology connected with the cloud causes a lot of parts covered and is capable to provide easy use and maintenance by cloud service providers also they build IT experts around the world to help the organisation to create the infrastructure according to the required web services (Kumar et al., 2019). So, the organisation moves to cloud solutions rather than investing the money in on-premise (OP) infrastructure, a part of OP, the organisation pays the SME, maintenance of infrastructure as well as taking more risk so that the cloud changes the IT industry and provides relief to the organisation, In cloud, in just some clicks, you can create the whole infrastructure according to organisation needs. Like DevOps, SMEs create the code (infrastructure as a service) in a file and build the infrastructure by using the Terraform utility (Terraform is an open-source tool) (Laghari et al., 2018). The cloud is a more popular aromatic buzzword in the IT infrastructure world. It has revolutionised the approach for company businesses, the connection also shared the information, cooperates, and administrates the computing web resources. According to the occurrence of the internet, cloud computing has revealed one more time to doing business also has empowered companies to standup above mentioned conventional OP IT infrastructure.

In the world, most cloud service distributor provides equivalent services, while the case of Asia trade is workable by AWS, AZURE and GCP. For that, it swiftly exchanges views of cloud web services promote. We could not forecast big suffering to close in the future (Walters, 2012; Nazir et al., 2020).

It is difficult to choose the best cloud service provider for the organisation due to multiple competitor service provider in the market, each service providers have different web services and their cost usage, most of the top three in the market AWS is taking high business and also provide best of best service but little bit costly as compared others (Laghari et al., 2019a, 2022). Azure is a Microsoft product and the second most popular cloud service provider but their graphical user interface (GUI) console is not as user-friendly as compared to AWS (Mishra et al., 2022a). The third GCP service provider also rapidly generate the revenue in cloud market as well as introduce new web service according to the market need, rest of the cloud service provider also in the market to provide web services, like Ali-baba, Rackspace, Huawei Cloud, and so on in working in the market (Kunz et al., 2022). Organisations choose the cloud service provider which provides cost-effective, vast guides available and great support available in case of clients needs help in configuration or troubleshooting any problem (Ghosh et al., 2014).

To best deliver the characteristics and interrelationships between Google GCP, Microsoft AZURE and Amazon AWS, their web cloud services ratio begins with a look at the three-cloud vendor's commitments which guide the direction they may be taking in the subsequent. To inspect the currently offered web services by the cloud service provider, their pricing design, and the concessions available to customers (Whaiduzzaman et al., 2014; Chou, 2015).

In this article, you will find a deep dive of comparison and analysis of OP and cloud computing which help one to select the best CSP from the top three cloud providers (AWS, Azure, GCP) according to their organisation's needs moreover benefits of cloud as compared to OP and its key features also highlighted in this article.

2 On-premise and cloud

It is not surprising that cloud computing (CC) has grown in reputation awarding its proposal and guarantee to offer businesses newfound flexibility, which saves time as well as the cost of developing speed and also concentrates on scalabilities (Ali et al., 2023). further, OP software installed at IT companies purchased servers and hunting the firewall has a deep offering for businesses and want to still adequately meet your organisation's needs. Additionally, OP applications are secure and safe, giving organisations the ability to control and manage all resources deployed at the site while the cloud generally manages at its level (Sabiri and Benabbou, 2015). Skills IT SME experts always choose the cloud and agree that they need new cloud and SaaS applications rather than add to their OP and hardware system (infrastructure) to achieve their business objectives.

What are the reasons OP exists in the business?

Quite clearly, the organisation could not replace the environment easily, especially when they drive important business processes in an organisation (Makita et al., 2019). The possibility of changing the status of the existing condition can be too high when you consider the possibility of losing or corrupting key data. In addition, at the middleware layer application often deficiency technical requirements, making it hard to develop a new system with identical functionality. Even if it is usually cheaper to migrate an old system in the long term, many solutions allow continued use (Teng et al., 2010; Velte et al., 2010). However, Legacy software should not be kept indefinitely as it involves certain risks.

2.1 Past of cloud compute

Structured to Technology analysis, the term CC was first cited in a 1996 Compaq central document. 1999 was a turning point for CC, one of the first companies Salesforce which deliver the enterprise application on the internet. Also, the starting point of (SaaS) is 'software-as-a-service' (Iqbal et al., 2016; Saini et al., 2019). Within 2002, Amazon introduced Amazon Web Services (AWS), another major development in CC. Its apartment of cloud-based web services included computation, storage, and even human common sense. Amazon launched the Elastic Compute Cloud (EC2), which authorises both businesses and individuals to charge VM virtual machines and run their computing applications.

2009 was another huge achievement in CC when Google Workspace (now Google Workspace) began providing browser-based enterprise applications. That same year, Microsoft entered the CC phase with Microsoft Azure, and organisations such as Oracle and HP soon followed suit.

2.2 *On-premise benefits*

Total cost of ownership (TCO), because you only pay once for your user license, an OP architecture solution can have a lower total cost of ownership ‘TCO’ than a cloud solution. Complete supremacy of your data, software, hardware, middleware, and application all products of yours. The company decides on upgrades, system changes, and configuration (Bibi et al., 2012; Johnson et al., 2019). System uptime with OP systems, you do not have to rely on internet connections or external factors to access your software (Pahl et al., 2013).

2.3 *Disadvantages*

- Large reserve spending: OP architecture typically needs huge upfront investments, which means ‘CAPEX’ capital expenditures are usually required. Additionally, you need to factor in maintenance costs to ensure support and feature upgrades (Pahl et al., 2013).
- Maintenance responsibility: in a premise environment, you are responsible for maintaining the server’s hardware, software, storage, backup, and DR ‘disaster recovery’ (Whitehouse and Buffington, 2012). That would be a problem for micro-organisation that has a limited reserve and sound technical resources.
- Extensive deployment period: OP deployments, configuration, and installations take too much time to be completed on servers as well as company users’ laptops and desktop systems. According to the OP details mentioned early (Ratametha and Veeragandham, 2009; Resende et al., 2021). So, for that, the overall organisation depends on their requirement for an architecture solution.

3 **Deployment OP**

In the OP environment, the software is deployed internally and within the company’s IT environment. A company is responsible for maintaining the resources and all connected equipment (Iyamu and Mphahlele, 2014).

CC: the various flavour for CC like private cloud, public cloud, and hybrid cloud. Over all resources hosted on the CSP premises, meant on their onsite PC computing environment, but users are accessing as well as using their services as much as they need to use it at any time.

3.1 *Cost OP*

Companies installed software at OP and they are responsible for the running software charges as well as for power utilisation onsite. Cloud computing companies choose a web-cloud computing model and only pay for their web services resources as they are used in seconds, minutes, and long-term investment, cloud provides free maintenance, and costs, as well as the price, adjusts either up or down based on consumption (Mtebe and Raisamo, 2014).

3.2 Control OP

At OP architecture, all data of organisations stored and kept onsite also full control of their running infrastructure, so for this reason multiple companies are switching the onsite environment to a cloud solution for data protection (Pahl et al., 2013).

CC: at CC architecture the joint vendors related to the data store work to gather and provide the best services as well as provide the best algorithm to secure your data which store in the cloud. CSP cloud service providers have multiple availability zones you deploy your service or Compute resources in multiple availability zones to avoid downtime case if one availability zone is affected your service can run from another zone. Availability zone comes under the region. Each region has multiple availability zones.

3.3 Security OP

Organisation that has particularly confidential information, such as public authorities and banks, example, need the exact level of security and data protection that an OP environment offers. A part of security is a major issue for all industries, so an OP environment makes confident customers but it's a high price to maintain the data (Ramachandra et al., 2017; Pahl et al., 2013).

CC: the security persists in CC deployment, cloud vulnerabilities are time by time published and patches are updated according to the security concern (Lamba et al., 2016), all patching is deployed by CSP, and everyone belongs to IT organisations having security concerns. Through customer data, details like bank usage user login ID/password credentials are hacked that why security threats are existent.

3.4 Compliance OP

One or more companies today regardless of industry, operate under some form of government control. Like 'HIPAA', there are several, plus the family 'FERPA', which provides records of students in detail as well as other government and industry regulations.

It is critical that they comply with the regulations and realised their data at all times at particular locations (Breiter and Naik, 2013).

CC: businesses always move to adopt a CC on their business need and always ensure that their vendor third party meets all the different regulatory demands in their industry (Mishra et al., 2022b; Berlato et al., 2020). Critical data should be protected as well as the privacy between partners, customers also employees should be guaranteed.

The details about all cloud instances size, GCP, AWS, AZURE and RACKSPACE services are given in Tables 1 and 2.

Additionally, to the barriers to creating a meaningful cloud web service cost balancing, all top three web service providers CSP have variable costs for web services based on server geo-location we can say that is a region so CSP billing for upfront payment, executed use, perpetual use, or off-premises use the rush hours (Rashid and Chaturvedi, 2019). Let us check Table 1, it mentioned the details and guidelines for savings the revenue of the organisation as well as getting discounted price points according to the long-term use policy so we can choose the services according to the organisation's need and saving the cost of company (Barker et al., 2015; Qi et al., 2017).

All useful web services are mentioned in Table 1.

Table 1 Comparison web-services provided by cloud organisations

<i>Service</i>	<i>AWS</i>	<i>Azure</i>	<i>GCP</i>
Compute	Elastic cloud compute	Virtual machine	Compute engine
App hosting	Elastic Beanstalk	Cloud services	App engine
Serverless computing	AWS Lambda	Azure Functions	Cloud functions
Container support	ECS/EKS containers	AKS container	Kubernetes engine
File storage	S3 storage service	Azure storage	Cloud storage
Block storage	Elastic block storage	Azure Blob	Persistent disc
Backup options	AWS Glacier	Azure backup	Cloud storage
Data orchestration	Data pipeline	Data factory	Cloud dataflow
Data Management	AWS Redshift	SQL data warehouse	Google Big Query
NoSQL Database	DynamoDB	Cosmos DB	Cloud Datastore

Table 2 Comparison between the All mentioned CSP instances cost with discount and Length of usage

<i>Provider</i>	<i>AWS</i>	<i>Azure</i>	<i>GCP</i>
Type of discount	Reserved instances	Reserved instances	Sustained use (SU) Committed use (CU)
Length of commitment	One to three years	One to three years	SU – no commitment
Discount available	Up to 75%	Up to 72%	SU – up to 30% CU – Up to 55%
Other discounts	Spot instances	Enterprise agreements	Preemptible VMs

The top CSP gives a discount on long-term usage with the length as mentioned in Table 2, one of the major discounts in AWS which are spot instance, this discount is useful for an organisation that wants to QA UAT for the architecture running the major application (Gupta et al., 2021). Suppose one organisation wants to use big data and test the 4,000 concurrent users then they will look at spot instances for testing to save the cost for temporary usage of systems, while so for OP the company purchase hardware, and if it is not loads testing failed and not fill the requirement then it huge loss of the company. Table 2 mentioned the length of time and discount in percentage. The advanced web-services technology contains many services and web resources that become ready for customers, while the terms of multiple CSP organisations gradually create web services according to the need of the market (Rani and Ranjan, 2014). Note price is variable time by time due to CPS policy.

As cloud technology is a trending top for mini businesses to worldwide corporations but it's a remnant enough concept that enfolds many online regions, while users start the transition for organisation business on the cloud for application delivery or infrastructure deployment. The organisation should check and well understand the dissimilarity and benefits of cloud web services with CSP. Although a lot of web services are expanding day by day, typically three cloud web service models that can be compared (Rani and Ranjan, 2014): examples of PaaS, SaaS and IaaS.

- *PaaS*: CSP web services related to platform are called platform-as-a-services, web services software is used mainly for applications. PaaS delivers a sub-structure for developers so that they can create customised applications. All data storage, network, and running servers can be controlled by the vendor and also the developer manages the applications (Laghari et al., 2021).
- *SaaS*: it represents cloud application services, which proceed widely used options for businesses in the cloud era (Free et al., 2015).
- *IaaS*: infrastructure-as-a-service, Infrastructure in the cloud is known as infrastructure-as-a-service and it is made of highly auto-scale and automated web services-compute resources according to the IT industry needs infrastructure in the cloud is automated and at any time we can build the whole infrastructure within a single click, it reused the model (Laghari et al., 2019b) and tool to create the script for automation like, Chef, Python scripting, Terraform, and transformation (Jamsa, 2022; Bhardwaj et al., 2010). So, organisations allow the business to purchase required resources and auto-scale when its need instead of buying hardware. These services are mentioned in Table 3.

Table 3 Could the service provider provide the SaaS, PaaS and IaaS

<i>Platform types</i>	<i>Common examples</i>
SaaS	Google Workspace, Dropbox, Salesforce, Cisco, WebEx, Concur, GoToMeeting
PaaS	AWS Elastic Beanstalk, Windows Azure, Heroku, Force.com, Google App Engine, Apache Stratos, OpenShifts.
IaaS	Digital Ocean, Linode, Rackspace, Amazon Web Services (AWS), Cisco MetaPod, Microsoft Azure, Google Compute Engine (GCP)

4 Types of cloud computing

Top CSPs in the market, including AWS, GSP and AZURE are spread in the market and provide the best web services and well studies about the market need and invent the services according to the market demand, this subsection provides us a clear idea about different types of cloud computing (Furht, 2010).

4.1 Public cloud

The standard cloud model used for public computing is called a public cloud, CSP creates resources like applications, infrastructure, and data storage which are available for customers as well as businesses over the internet (Hofmann and Woods, 2010; Laghari et al., 2017, 2019b). The top CSPs like Microsoft, Google, and Amazon have their infrastructure at their data centre in multiple regions, CSPs also rapidly increase the business and fascinate the user with launches the new web services in the market and all user use will be only by internet connection (Rajan, 2018). One more thing is that the public cloud is no direct connection to your on-premises data centre, users only connected public cloud architecture to the internet (Ren et al., 2012). CSPs also provide services that may be offered free email accounts like Gmail, and some web services may

be provided by pay as you use for businesses purpose (Zwattendorfer and Tauber, 2013). Customers have limited visibility as well as control over it where the computing infrastructure is hosted in the region. PC web services are too much easy to administer as well as cost-effective, note it is not considered as secure as a private cloud.

4.2 Private cloud

Building and creating the infrastructure as well as software for the organisation at an onsite physical location is called a private cloud (Garg et al., 2022). Most organisations are reluctant to use the public cloud due to security reasons that's why user keep their environments onsite to save security as well as control customised applications so a PC is preferable. Overall infrastructure and services are maintained by self, PC is generally used by corporate buildings as well as the house. So, for cost in PC, is more costly than public cloud cause once you purchase the hardware for network data storage then it's your responsibility to maintain it as well as keep secure data transfer from one hub to another hub (Jennings and Stadler, 2015).

4.3 Hybrid cloud

Most finance business organisations maintain different e-business applications along with different levels of security so for that, they used hybrid cloud services, which are a combination of public and private clouds as mentioned in the above details, this type of cloud service is implemented by different CS providers (Kautish et al., 2022). One of the major disadvantages of these services is that we have to manage different security platforms together (Vaishnnave et al., 2019).

5 Data privacy

Sharing cloud infra environment could lead to confidence and privacy issues, as well as disclosure and correlate storage with conflicting outcomes for the legitimate status as well as defence of personal customer and their business data. The region-wise location of the cloud data affects the data-protection function for storage and processing; legitimate consequences could arise if data is residing in several places like multiple hosted regions around the world at the same time (Wenge et al., 2014).

If data is stored inside, it increases the vulnerabilities to moving the business data, copying, and accessing. threats could come from inside users like malicious cloud provider users, customer users, and malicious cloud or malicious vendor users as well as business-data leaks occur due to the weak privileges and loss of security access privileges across multiple domains hosted also the failure of physical transport systems to carry data for cloud and backup as well as electronic backup data control.

The cloud solution makes protecting data and enforcing privacy, cybercrime and identity theft, more difficult because the companies do not have direct rules over the data implemented through a CSP. Data sharing resources with a vendor like a customer and a web services cloud expose data that can be confiscated if multiple organisations has broken the law.

6 Cloud services and data availability

Disaster recovery (DR) methods and UAT are essential for the consequences of a disaster to ensure the using web service and data availability, it's the best practice in the finance sector which implement the DR in case a disaster occurs (Perri et al., 2022). Second thing is the risk that containing secure business data could, so CSP prevents the testing of data backup recovery methods during the event of an incident, while most cloud users may be given high value in recovery activities (Bouchenak et al., 2013). As the Internet is the main medium for processing and transmitting data as well as it creates availability issues due to potential connectivity also bandwidth speed limitations.

6.1 Data integrity

Most vulnerabilities threaten is the integrity of the application, network, DB, and system software on the shared web-cloud environment with global access if the latest patch does not apply appropriately as well as on the time which is recommended by CSP. One more risk in the web-cloud computes environment are unapproved changes to the data and systems by the SP, which could compromise the availability as well as the integrity of data. Web-cloud hosting environments pose a data integrity threat if system resources are not productively sorted out between customers (Zafar et al., 2017).

6.2 Data encryption

The major risk in web services cloud computing is insufficient data encrypts as well as RSA-key management. The web service cloud environments are shared with multi-tenants also CSP has granted access to the business data as well as posing a risk of data, which unauthorised access on hosted cloud computing (Yang et al., 2020). So, regulated and critical business data is transmitted over a cloud network which increases the risk of adoption, hijacking, as well as leaking the data information from disks also the live environment is open for malicious co-tenant or malicious cloud SP. CSP introduced a few encrypted algorithms for transferring data over the internet, so all data is encrypted and decrypted according to the algorithm.

6.3 Virtual/logical access

Root access (administrator) provide through the internet while OP all administrative privileges are controlled and restricted connection by the internal network, so the risk of unauthorised as well as insufficient permission access, such as super user access, increases when data is processed outside the company which mean that outsourced service provider can bypass control over internal programs.

Business data access through the internet involves more vulnerability as well as more risks for the organisation due to security breaches (Ahmad et al., 2022). Access the business data or cloud-web applications that are accessible worldwide through the cloud as well as shared with other users due to the multi-tenancy of EC2 cloud-compute like weak cipher authentication mechanism could be insecure user behaviour, for example, either password reuse or weak password which is the inherent limitation of the single-factor authentication mechanism and inadequate segregation of duties. The migration of infrastructure leads to potentially unauthorised access and disclosure,

including challenges such as confidential user data and user credentials so it should be strong authentication like two multi-factor authentications by email or cell phone code authentication, trust management, and delegated authentication across all types of cloud web services (Yamato et al., 2015).

6.4 Network security (NWS)

In a cloud environment, all users are worried about hacking and intrusion. Incursion and Harm risks include the attacker getting access to the application as well as business data via some kind of remote access system, i.e., ssh, RDP (Nazir et al., 2021). The cloud-web application injection vulnerabilities are exploited by manipulating input service in such a way that portions of the input are exposed as executed code to the interpreted according to the programmer's intent for example cross-site scripting, SQL, command injection as well as security threats such as hidden hackers, man in the middle attack, sides channel attack, social media networking attack, CLI mode attack and one of the major denial of service (DoS). DOS attack is also enhanced with DDOS attacks which are 10 times faster than DoS, authentication attacks, and important threats in the web cloud computing Environment (Waqas et al., 2022; Khan et al., 2022b; Khan et al., 2022c).

The corporate boundary is vaporised in the cloud computing environment, therefore, the bottom advisor affects everyone's security. The corporate firewall form is best for security policies as well as for network security which is either no longer accessible in a cloud solution or those policies are no longer under the control of the owner resource, while it is the responsibility of the CSP.

The cell phone is a new risk for cloud-enabled users who can access the services as well as the business data transmitting them across the corporate network so it leading the security venerable (Fisher, 2018).

6.5 Physical access

If an organisation globally accesses a huge amount of data either put down or get through the web cloud so the organisation opens multiple distributed threats attacker can get access from one virtual location rather than a secure onsite location (Achar, 2022; Khan et al., 2022a, 2022d, 2022e).

6.6 Compliance

The organisation always makes sure to act by a set of requirements by their industry or government body for securing both external and internal business information data as well as running multiple business applications. In cloud computing, all servers, applications, and data are hosted at an offsite location. Compliance helps to protract your organisation from threats and secure your infrastructure and keep safe from lawsuits and fines. Compliance with some laws and regulations like GLBA Act, HIPAA Act, PCI-DSS restriction applicable to storing data in transit, SOX Act, SAS70, and ISO (Guo and Song, 2010).

CC: CC growth forecasts indicate significant implementations and developments of CC web services to make cloud infrastructure more resilient and secure, appropriate controls and weakening of the security risk should be executed. Risk should be considered to ensure the target completeness, availability of applications, and integrity of

business data on any cloud, mostly suggested several controls that might be considered to mitigate CC security risks included data management, security control, logical access, physical security, network security, compliance, and virtualisation.

Before jump into characteristics of cloud computing, let me introduce another computing which is called fog computing.

Fog computing: fog is an additional layer in a more complex system, in simple words a data processing and storage system. Additional expenses to one organisation budget. Enterprises need to buy edge devices: routers, hubs, gateways. Limited scalability: fog does not scale like a cloud.

7 Fog and cloud computing

Fog computing is a decentralised computing infrastructure or process in which computing resources reside between a data source and a cloud or other data centre. Fog computing is a paradigm that provides services to user requests on edge networks. Devices on the fog layer typically perform network-related operations, such as routers, gateways, bridges, and hubs. Researchers envision these devices performing both computing and networking tasks simultaneously. Although these tools are resource-constrained compared to cloud servers, their geological spread and decentralised nature help provide reliable services with coverage over a large area. Fog is the physical location of computing devices that are much closer to users than cloud servers.

7.1 Security challenges in fog computing

Fog computing is a distributed computing paradigm that extends cloud computing capabilities to the edge of the network, bringing computing resources closer to the end-users and devices. While fog computing offers several benefits, such as reduced latency, improved network efficiency, and better quality of service, it also brings along some security challenges. Here are some recent security challenges faced in fog computing:

- *Data security:* fog computing involves processing and storing data at the edge of the network, which may increase the risk of data breaches and unauthorised access. Data encryption, access control, and secure data storage are crucial to ensure data security in fog computing environments.
- *Network security:* fog computing relies on a complex network infrastructure involving multiple devices and systems, which may be vulnerable to cyber-attacks such as denial-of-service (DoS) attacks, man-in-the-middle (MitM) attacks, and network intrusion. Network security mechanisms such as firewalls, intrusion detection/prevention systems (IDS/IPS), and virtual private networks (VPNs) can help protect fog computing network
- *Device security:* fog computing involves a large number of interconnected devices, including sensors, gateways, and edge servers, which may be vulnerable to attacks such as malware, phishing, and social engineering. Device security measures such as authentication, secure boot, and firmware updates can help mitigate these risks.

- *Privacy concerns*: fog computing involves collecting and processing large amounts of data, which may include sensitive information about individuals or organisations. Data privacy laws such as GDPR and CCPA impose strict regulations on data handling and storage, and fog computing providers need to ensure compliance with these regulations.
- *Lack of standardisation*: fog computing is a relatively new technology, and there is a lack of standardisation in terms of security protocols and mechanisms. This makes it difficult to ensure interoperability and compatibility between different fog computing platforms and devices.

In short fog computing brings along several security challenges, including data security, network security, device security, privacy concerns, and lack of standardisation. Addressing these challenges requires a comprehensive approach that involves a combination of technical, organisational, and regulatory measures.

7.2 Insider attacks in fog computing

Insider attacks are a type of cyberattack that are carried out by individuals who have authorised access to a system or network. These individuals may be employees, contractors, or other authorised personnel who have access to sensitive information or critical systems. In the context of fog computing, insider attacks can pose a significant threat, as fog servers may contain sensitive data and resources that can be exploited by insiders.

Following are some examples of insider attacks that can occur in a fog computing environment:

- *Data theft*: an insider with authorised access to a fog server may steal sensitive data, such as customer information or intellectual property, and sell it to a third party.
- *Sabotage*: an insider with access to fog servers may intentionally disrupt the operation of the network or cause damage to the infrastructure.
- *Malware injection*: an insider may install malware on a fog server, which can be used to steal data, disrupt the network, or launch other types of attacks.
- *Unauthorised access*: an insider may use their access privileges to gain unauthorised access to resources or systems that they are not authorised to use.
- *Misuse of privileges*: an insider may misuse their privileges to perform actions that they are not authorised to perform, such as accessing sensitive data or making unauthorised changes to the network configuration.

Followings are crucial characteristics of CC:

- demand self-service
- resource pooling
- economical
- availability and resiliency
- large network access

- easy maintenance
- measured and reported service
- rapid elasticity and scalability
- security
- automation.
- *On-demand and self-service*: one of the essential as well as major features of CC is, it allows 24-by-7 continuous monitor the web services and virtual system, storage, and network, that is a major feature of CC that customer can control the computing power according to their needs.
- *Resource pooling*: this is an important characteristic of CC; we can say in simple words that a CSP can share the resources among multiple customers to provide each with a different set of services according to the organisation's needs. This is a multi-client plan of action that can be applied to processing the web services, bandwidth, and data storage services. The management process which assigns the resources in real-time is not at odds with the customer's experience (Carroll et al., 2011).
- *Easy maintenance*: ease of maintenance is the beauty of the cloud, even servers' effortlessly downtime remains minimal and maintained sometimes absolutely zero (Song and Sohn, 2022). Cloud computer-based resources often undergo a few updates to optimise their prospective as well as ability. The updates are more usable with the devices and faster than the previous version on CC. The cloud property allows workloads that require large numbers of servers to be run inexpensively but only for a small period. Customers have such workloads that can be operated very cost-effectively due to the rapid scalability of CC (Arockiam et al., 2017).
- *Economical*: CC help in decreasing the IT expenses of all type of organisation. With CC the customer has to pay the administration for the storage space they used. There are no hidden or additional charges to be paid. Management of cost got easier through the dedicated cost calculator provided by CSP.
- *Measured/reporting service*: the reporting service is cloud features that make them the best choice for businesses. The measurement and reporting service is useful for both CSPs and their customers. These web services allow to provider and the customer to monitor and report the web services, which have been used in the organisation and also assist to monitor the billing and ensure optimal resource utilisation.
- *Security*: security is one of the best features of CC, web services have multiple availability zones at every region where backups are created at a time automatically as well as manually, it is a copy of the stored data to prevent data loss. For example, if one EC2 instance accidentally loses the data, a copy of the version will be restored from the availability zone in a single click, this is the beauty and best feature that comes in handy when different users are working on a specific file in real-time and a file suddenly becomes damages.

- *Automation*: in CC, automation is a key feature, which automatically installs, configures, and maintains a cloud web service is called automation. Automation is the process of making the most of technology and reducing the manual effort to create the infrastructure and network. However, achieving automation in the cloud ecosystem is not easy. It requires the installation and deployment of a virtual machine, server, and large storage, once successfully deployed all resources then the organisation keeps constant maintenance (Givvehchi et al., 2013).
- *Resilience*: this is another beauty of CC; resiliency meant the potentiality of the service to quickly recover from an interruption. Cloud resiliency is measured that how quickly its databases, servers, and network systems restart and recover from any disaster (Jhawar and Piuri, 2017). Availability is another important feature of CC because cloud web services can be accessed remotely, and there is no geographic restriction and limitation in cloud resource usage.

7.3 Wide network access

The customer can access the multi-cloud business data from any location with just one device through an internet connection as well as transfer the business data to the cloud. IT SME engineers can access the console of CSP easily from anywhere through the internet. CSP saves network usage by monitoring as well as guaranteeing the various web service measurements that reflect how clients access the cloud resources concerning time, latency, data throughput, and all of the available web services. CC and OP can furnish your e-business/business according to the IT infrastructure required (Malathi, 2011). The lookalike you pick will likely turn on the level of security you require to meet the compliance standards as well as the price structure your organisation prefers. Table 4 described the business need where we can easily find the availability either on physical server or cloud CSP. Some of available on both side like secure data highly encrypted and few are in the cloud and onsite (physical location) In the cloud one of the best business need available if an organisation need automated backup and recover as well as automation, suppose organisation needs like to like environment immediately so cloud have function where administrator just create the script for building the like to like environment, it will be on single click the all over services created according to the business need, while at on-premises it's too hard to create like to like configuration and build the architecture. Large investment taking at onsite while in a cloud company only paid as they used the services. Also, it is very easy to scale the environment but, onsite you will first get the approval for a resource to add to the server, then it will be taking too much time to reach after approval and purchase but in the cloud, it on few clicks the resource will available according to the need business need (Boillat and Legner, 2013). Worldwide Enterprise Infrastructure Buyer and Cloud Deployment Forecast, 2019 to 2025 (SPENDING) are given in Figure 1.

Figure 1 is overall usage and forecasted statistics from 2019–2025 of Consumers having or shifting their IT infrastructure on on-prem, hybrid cloud, and cloud. In 2019 51.6% of IT Infrastructure resides on-prem but you can see in the above figure, its graph will drastically decline to 33.9% till 2025.

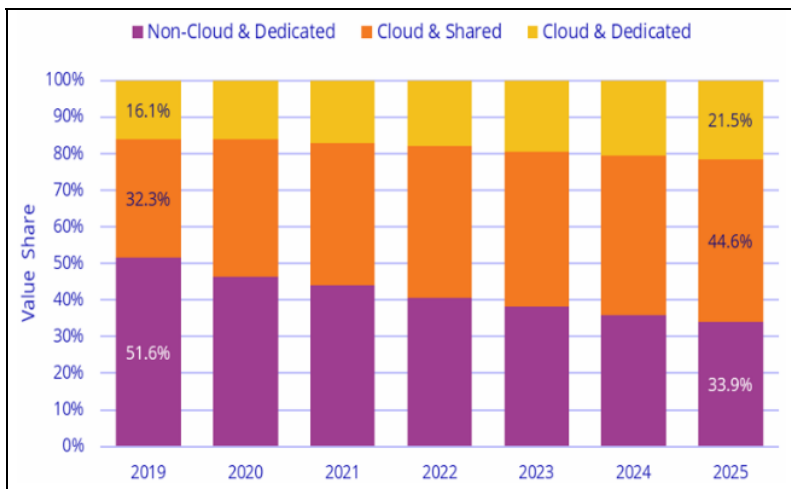
Reporting to the International Data Corporation (IDC), this firm conduct worldwide quarterly infrastructure tracker for enterprise; consumer spending, and deployments of cloud on compute and storage infrastructure products for cloud infrastructure,

including shared environments and dedicated, which is continuously increasing yearly by 12.5%, over year in the first quarter of to \$15.1 billion in 2021 (1Q21). Investment of infrastructure for non-cloud growth is 6.3% yearly in 1Q21 to \$13.5 billion (<https://www.sciencedirect.com/science/article/abs/pii/S1084804517301613>; <https://www.idc.com/getdoc.jsp?containerId=prUS48050621>).

Table 4 Model compared of cloud and on-premise

BUSINESS NEED: data centre single tenancy	Public cloud (CSP provide)	On-site (physical location)
BUSINESS NEED: secure data with highly encryption	Public cloud (PC)	On-site
BUSINESS NEED: zero downtime risk	Public cloud (PC)	On-site
BUSINESS NEED: automation, data backups and recovery	Public cloud (PC)	On-site
BUSINESS NEED: control and complete data visibility	Public cloud (PC)	On-site
BUSINESS NEED: resource usage price, pay as you go	Public cloud (PC)	On-site
BUSINESS NEED: regular investments infrastructure requires large	Public cloud (PC)	On-site
BUSINESS NEED: scalability, easy to scale up and down	Public cloud (PC)	On-site

Figure 1 Cloud deployment (see online version for colours)



As the world gradually begins to head up and the business market from the COVID-19 pandemic, the economic sectors that suffered the biggest losses in revenue are reversing to heightening. A long-term effect on the Infrastructure industry will be an increasing factor on web cloud platforms (CP) for the delivery of social, educational, and commercial applications, as well as improve focus by industries on the stability of

business and management of risks, helps to drive digitally transformed start-ups and to enhance the use of infrastructure-as-a-service models (IaaS).

8 Cloud solutions vs. on-premise solutions survey

A survey was conducted on cloud vs. on-premise solutions from multiple organisations where subject matter experts (SMEs) participate and gave feedback about the on-premise and cloud, Figure 2 is the graph showing how many organisations participated in this survey as well as the number of SMEs provided their feedback on survey conducted by the authors. Twenty-one organisations participated in this survey and from each organisation, one SME provided its feedback except Royal Cyber where three SMEs provided their feedback on this survey.

Figure 2 Research conducted by an organisation (see online version for colours)

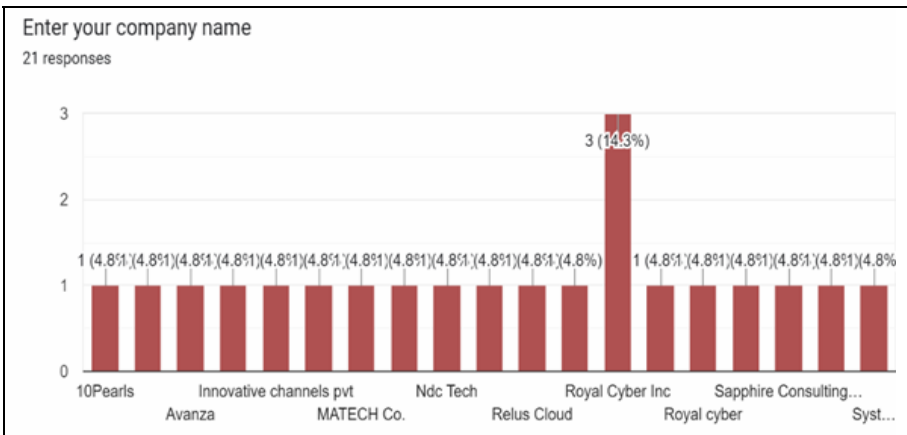
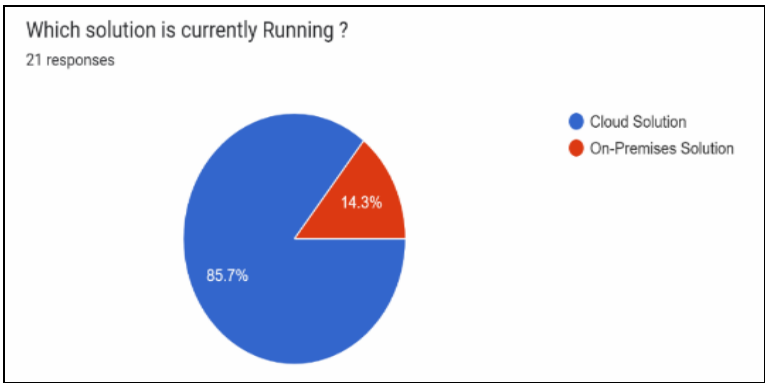


Figure 3 Solution currently running (see online version for colours)

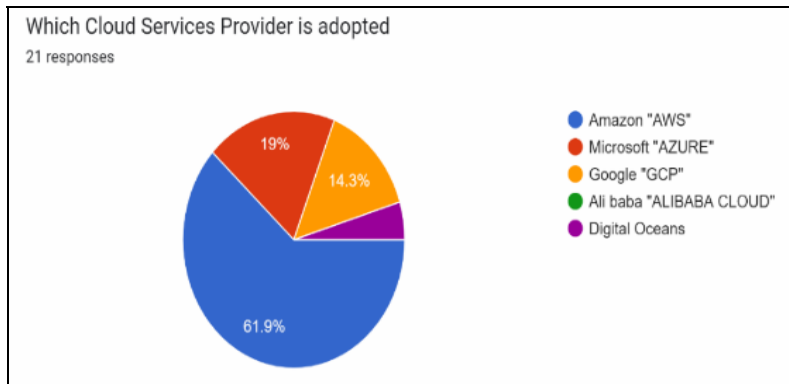


Around 21 organisations have feedback about the cloud and on-premises, most of the companies using cloud solutions, which are mentioned in Figure 3. There is an 85.7% graph of cloud solutions where on-premises solutions run 14.3 % out of 100% which

seems clear that most organisations like cloud solutions due to many reasons, major reasons are cost saving due to maintenance as well as scale the environment at any time.

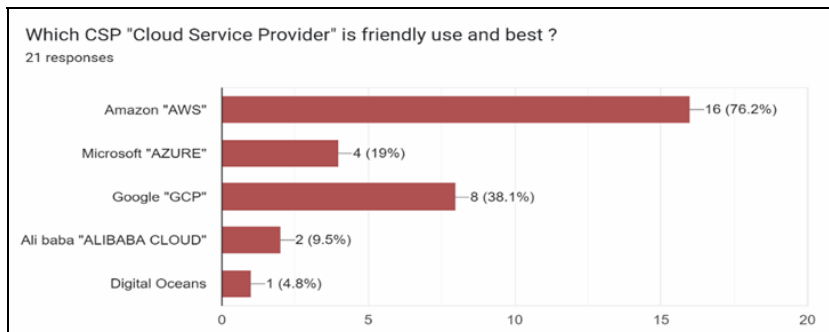
So far cloud solution providers, have everyone choose according to the web services which is compatible with their environment, but if you check Figure 4, AWS is top of them, which means AWS provides the best web services as well as the best maintenance around in the world where they have data centre the regions. 61.9% company use AWS while in the second number Azure is in the market.

Figure 4 Adoption of cloud services (see online version for colours)



In Figure 5, we collected customer feedback about CSP, why and which CPS is friendly to use, again a higher rate of AWS cloud in the market, which is adopted due to friendly use and easy access to the AWS console where the system administrator manages the resources. 76.2% of customers adopted the AWS while GCP is on second rating for ease to use resources via GCP console.

Figure 5 Cloud service provider CSP (see online version for colours)



Why did the organisation suddenly switch their IT infrastructure, Figure 6 rated the beauty of scalability, availability, and pay-as-you-use-the-web resources that's why organisation move their on-premises to cloud solutions. CSP provides unlimited usage and resources according to the business need.

Figure 6 Cloud services vs. on-premise (see online version for colours)

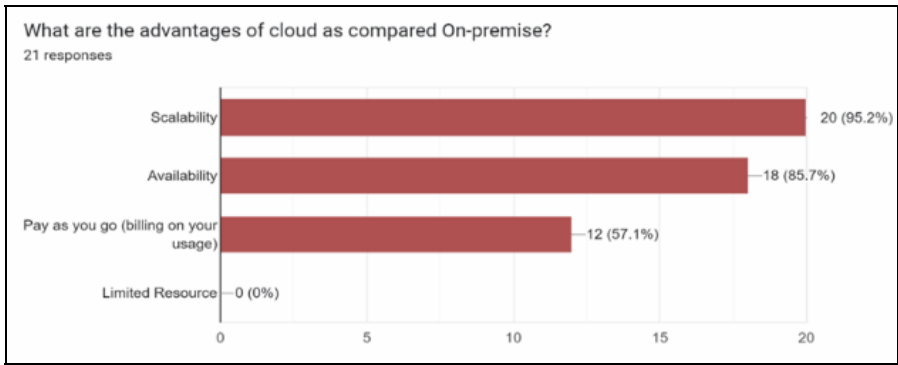


Figure 7 shows most of the company lose their annual capital due to maintenance cost, long-term investment on on-premise as well as employee hard to monitor all resources and definitely if the company hire more candidate who manages and monitor their running systems so it affected the company revenue that's why the company save the cost and switch the on-premise to cloud.

Figure 7 Disadvantages of OP over cloud (see online version for colours)

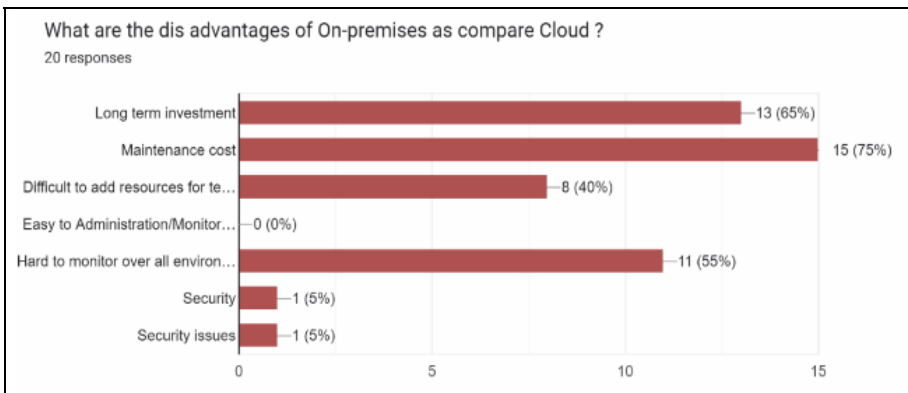


Figure 8 Issues of on-premise compared to cloud (see online version for colours)

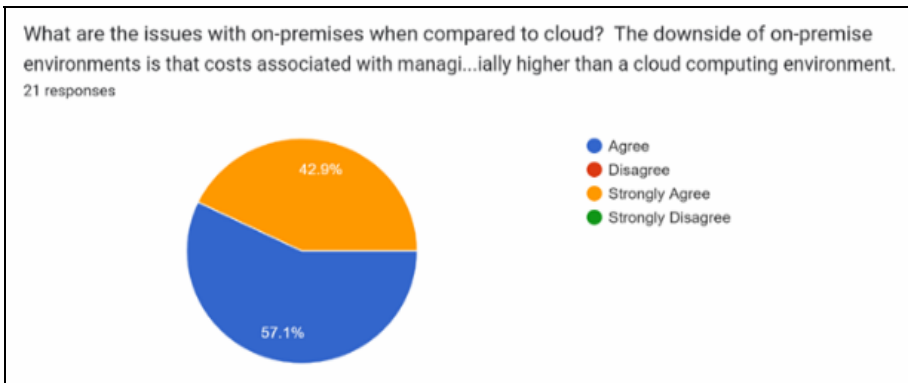
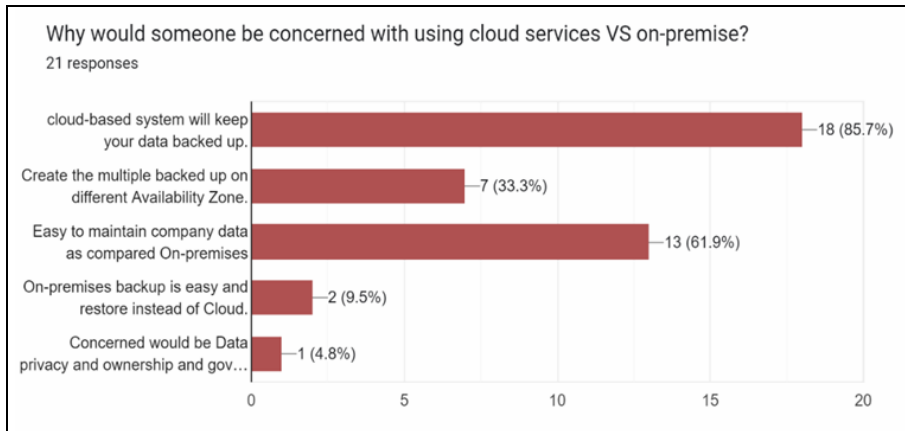


Figure 8, showing the overall organisation agreed with the statement about on-premise, it is a costly solution, but the organisation keeps its environment due to private data security, which is the main point they are on in the on-premise environment.

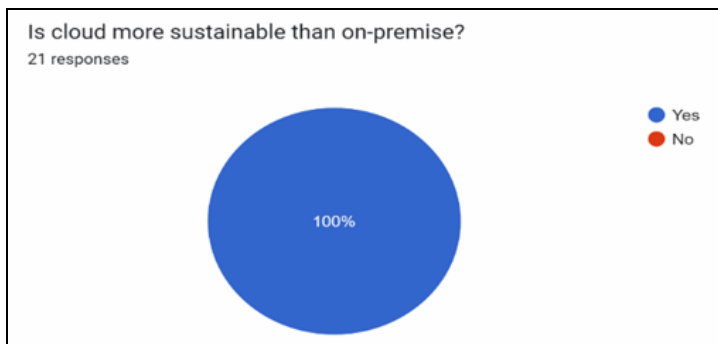
It is the major beauty of a cloud solution where your organisation keeps the backup of the data, which is an automatic backup on different AZ availability zone even, they purge backup with the help of web services according to the business need. Figure 9 represents the feedback in percentage on the backup as well as it is easy to maintain company data while it is too hard on-premise.

Figure 9 Response of cheaper on-premise and cloud (see online version for colours)



Again, the organisation agreed that the cloud is the best solution as compared to the OP, it is more sustainable than on-premises. Figure 10 shows 100% positive feedback on it.

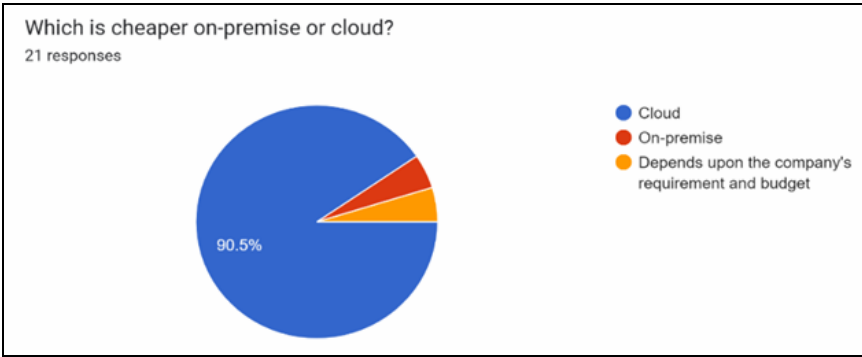
Figure 10 Cloud vs. on-premise suitability (see online version for colours)



As already discussed in the survey paper the cloud is cheaper than on-premises cause all CSP provider collects the charges against the web services which in the organisation used not for all idle resources, one of the best resources will be available on the spot, if organisation want to UAT for the production application which will be running life, they never go for long investment like on-premises they purchase the hardware for QA, the cloud provides the high resources to you for limited time usage then you can test for your

future development. Figure 11 shows 90.5% who agreed that the cloud is cheap and easy to use.

Figure 11 Productivity comparison of cloud and on-premise (see online version for colours)



The organisation adopted the cloud due to the best of the best web services it provided, the business rapidly scales its IT infrastructure during peak business like on holy days or special events they offer a 70% discount on products, that time organisation is not worried about the load to cause cloud solution to provide the auto-scaling according to the load. So far load during the peak time cloud auto-scales the infrastructure according to the business need while on-premises servers not sustained the load and businesses face the disturbance and loss of revenue. Figures 12 and 13 represent the positive feedback about a better environment.

It is the best example in Figure 13, if your organisation has an online e-store business application. The special event they announce the promotion of products so definitely a lot of traffic will search for and order products to get the benefits from the organisation so for that during peak user load the cloud solution provides the auto-scales your architecture and reduce automatically according to the usage. This is awesome beauty and organisations like that one for their business.

Figure 12 Comparison of better cloud and on-premise (see online version for colours)

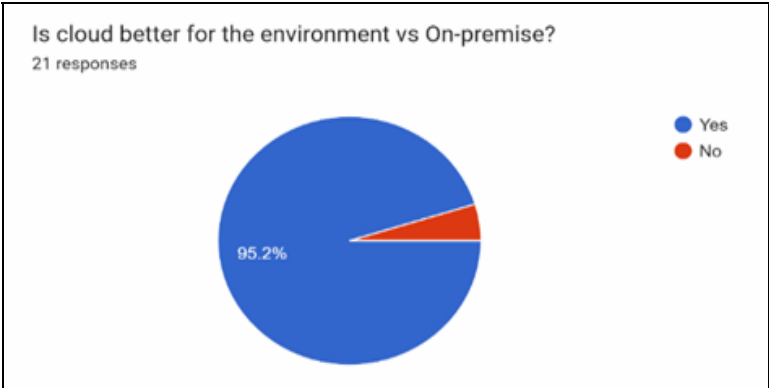
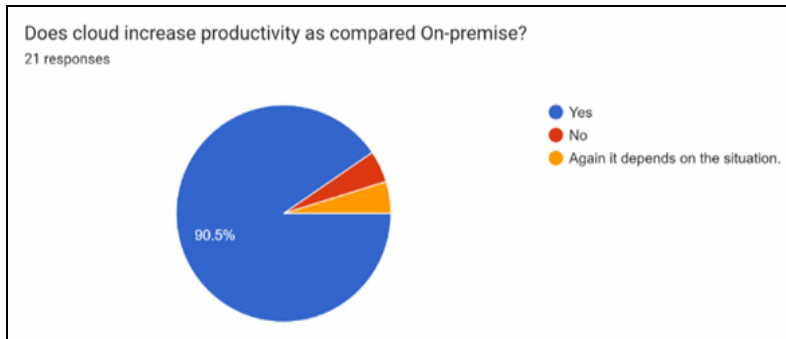
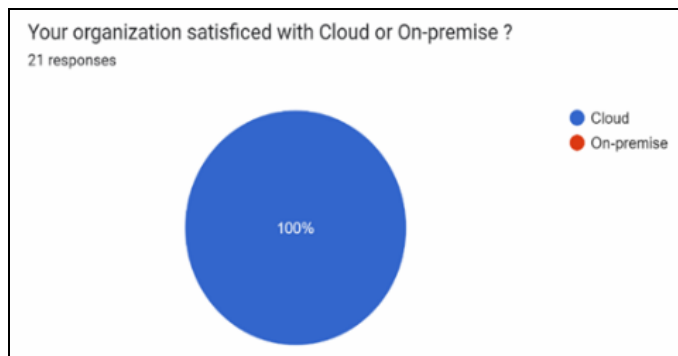
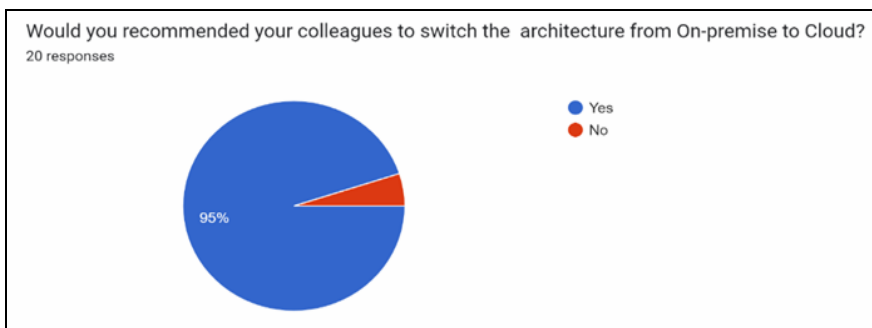


Figure 13 Cloud vs. on-premise productivity (see online version for colours)

So far satisfaction with cloud and on-premises is positive feedback on cloud solutions because they provide SaaS, PaaS and IaaS, these facilities seem good for business to move their Infrastructure and use CSP services and satisfied according to the business need, in Figure 14 got responses from 22, 21 is 100% satisfied to use the cloud solution.

Figure 14 Organisational satisfaction on cloud vs. premise (see online version for colours)**Figure 15** Cloud recommendations percentage to switch from on-premise (see online version for colours)

In a survey we asked the organisation's administrator about moving the on-site solution to the cloud, they recommended the next organisation move on-premises to the cloud for business and avoid the huge investment in resources and maintenance, nowadays overall

big organisations use the cloud to facilitate customer and providing high availability if a disaster occurs in any regions. The cloud provided backup restoration on each region, as well as a cache mechanism for social organisations, like Netflix, which uses the cloud for streaming for the customer, and edge center provides the cache for better performance. Figure 15 shows the 95% who recommended the cloud solution.

The followings are feedback from the organisation's administrative comments on the benefits of cloud and on-premise which run the IT infrastructure of the industry:

- Comments about on-premise and cloud solutions? Which one is best for the organisation and why we adopt it.
- Nowadays cloud solutions are everywhere and easy to use. Further, they are easy to manage and user-friendly accessible and controlled, secure, and within budget.
- Cloud service is better cheaper easy to maintain and monitor and facilitates integration between services. Where on-premise doesn't provide these facilities.
- The cloud solution is best due to its less maintenance cost.
- The cloud environment is secure less administrative, highly scalable, and accessible at lower latency to the end customer in the world. So, the cloud has more benefits if compare with on-prem.
- Better than on-premise due to its flexibility, reliability, and security, the cloud removes the hassle of maintaining and updating systems, allowing you to invest your time, money, and resources into fulfilling your core business strategies.
- It is very good to use, cloud is easy to use as well as cost-effective while on-premises is difficult to maintain and very hard to restore the backup, the cloud is fast as compared to on-premises.
- The cloud solution is the best because it is scalable and increases productivity.
- Cloud is the best.
- Using cloud for SaaS, PaaS and IaaS, while on-premises it is difficult to use these above services. Our company is satisfied to use the cloud services and save the maintenance cost.
- Overall cloud is better.
- Cloud is growing fast in the world due to the easy build of the infrastructure and you know well about on-premises it's hard to manage and very costly to maintain. Cloud is the best solution.
- Cloud solutions.
- Cloud is the best for me.
- Some of the questions in this survey form are not correctly inquired and lack clarity. Some need more than two options. Overall, the cloud may not be the best solution in every case. It all depends upon the corporate's requirements, budget, governance, and Data privacy concerns. Nevertheless, the cloud is not a bad solution either and has proven to be a lesser pain area than on-premise. Each solution depends on the context and need.

- The cloud solution is better than on-premises, cloud service providers provide a lot of web services which is the best option for our organisation's use.
- I recommended the cloud solution as the best option for any organisation.
- The cloud is the most popular nowadays because cloud service provider provides SaaS, PaaS, and IaaS while on-premises it is hard to manage all things so the cloud is the best solution for building quick Architecture in a single click.
- Cloud is hot in the market due to IaaS, PaaS, and SaaS, and on-premises the company paid for maintenance and installed the software with the help of an Administrator and monitor the architecture. so, cloud is best as compared to on-premises
- With the cloud, companies can add more resources as they grow and their requirements increase. In contrast, with on-premises, organisations usually have limited storage and computing power.

The organisation's administrative feedback about on-premises and cloud, the majority is using cloud solutions due to saving the cost of maintenance, easy to create infrastructure, and easy to scalable the resource according to the company's needs and the company paid only the charges which they use the web-services for short term, so far OP company paid a lot of charges for maintenance as well as purchase the hardware if want to scale the environment. 95.2% of the administrative of the organisation said that the cloud is best than on-premise while 100% positive feedback for the cloud solution. Cloud service providers: Amazon AWS is top-ranked in the market for their web services and easy-to-use all services, as an above figure of organisations 61.9% use the Amazon AWS (CSP) and 95% recommended moving on-premises to a cloud solution.

9 Conclusions

In this paper, we conducted a systematic analysis of cloud computing and on-premise services. Cloud computing has surged in popularity, with its appeal and promise of not just offering the latest flexibility for businesses, it saves money, save time as well as improve agility and scalability. So, far on OP infrastructure, the company or vendor is responsible for the installation of required software at the physical site as well as responsible for their firewall security. We cannot deny the fact that on-site or on-premise applications are secure and reliable like the banking sector Pakistan state bank SBC is not allowed to use the cloud due to public data. Now recent state bank advisory allowed us to use PTCL Cloud and the main reason we believe is the easiness of deployments and infrastructure as well as Security and backup policies offered in CSP.

Though IT upper management agrees that they need to use cloud and SaaS applications in addition to their on-site (on-premises) and legacy systems to achieve their business goals. This model is called a hybrid model. But still many organisations still stick to on-site solutions and not agreeing to use web cloud services. Both solutions come up with unique services as mentioned in Table 1 and key characteristics but after proper examination and your business needs you can determine and decide which solution would be a great fit for your company.

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